

Linear Regression

Technical Notes

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Linear regression is a statistical method used to model the relationship between a response variable and one or more predictors, or covariates. When there is a single predictor, the model is called **Simple Linear Regression**; when there is more than one predictor, it is called **Multiple Linear Regression**.

Linear regression is a fundamental statistical model and is also widely used as a supervised machine learning method for regression problems. It provides a natural starting point for studying regression, introducing many of the concepts that also appear in more sophisticated models.

1 Simple Linear Regression

Suppose we have a training dataset

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N, \quad (1)$$

where x_i is the predictor and y_i is the corresponding response for observation i . In tabular form,

y	x
y_1	x_1
y_2	x_2
$\vdots$	$\vdots$
y_i	x_i
$\vdots$	$\vdots$
y_N	x_N

Table 1: Training dataset for simple linear regression.

Suppose that the conditional mean of the response can reasonably be modeled as a linear function of the predictor. The simple linear regression model is

$$y_i = \theta_0 + \theta_1 x_i + \varepsilon_i, \quad (2)$$

where θ_0 is the **intercept**, θ_1 is the **slope**, and ε_i is an error term representing the part of y_i that is not explained by the linear relationship.

The conditional mean of the response is therefore

$$\mathbb{E}[Y \mid X = x] = \theta_0 + \theta_1 x. \quad (3)$$

The goal is to estimate the unknown parameters θ_0 and θ_1 from the observed data. Once estimates $\hat{\theta}_0$ and $\hat{\theta}_1$ have been obtained, the fitted regression function is

$$\hat{y}(x) = \hat{\theta}_0 + \hat{\theta}_1 x. \quad (4)$$

For observation i , the corresponding fitted value is

$$\hat{y}_i = \hat{y}(x_i) = \hat{\theta}_0 + \hat{\theta}_1 x_i. \quad (5)$$

For each observation, the difference between the observed response and its fitted value is called the **residual**:

$$e_i = y_i - \hat{y}_i = y_i - \hat{\theta}_0 - \hat{\theta}_1 x_i. \quad (6)$$

It is important to distinguish the residual e_i from the error term ε_i . The error term is defined with respect to the unknown population regression function,

$$\varepsilon_i = y_i - \theta_0 - \theta_1 x_i,$$

and is therefore unobserved because the true parameters θ_0 and θ_1 are unknown. In contrast, the residual e_i is computed after fitting the model and measures the discrepancy between the observed response y_i and its fitted value $\hat{y}_i$.

The question is then how to choose $\hat{\theta}_0$ and $\hat{\theta}_1$ so that the fitted regression line represents the observed data as well as possible. The standard approach is **Ordinary Least Squares**.

1.1 Ordinary Least Squares

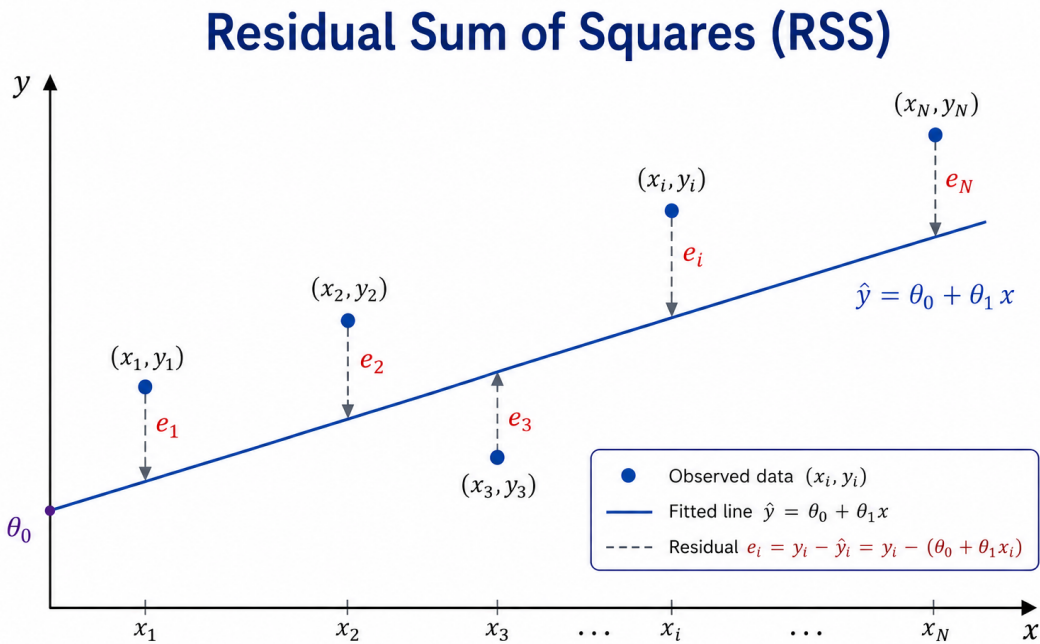


Figure 1: Residuals between the observed responses and the fitted regression line used to compute the Residual Sum of Squares (RSS).

1.1.1 Residual Sum of Squares

Ordinary Least Squares (OLS) estimates the model parameters by choosing the values of θ_0 and θ_1 that minimize the sum of the squared differences between the observed responses and the values predicted by the regression line.

For arbitrary parameter values θ_0 and θ_1 , the **Residual Sum of Squares (RSS)** is defined as

$$\begin{aligned} \text{RSS}(\theta_0, \theta_1) &= (y_1 - \theta_0 - \theta_1 x_1)^2 + (y_2 - \theta_0 - \theta_1 x_2)^2 + \dots + (y_N - \theta_0 - \theta_1 x_N)^2 \\ &= \sum_{i=1}^N (y_i - \theta_0 - \theta_1 x_i)^2. \end{aligned} \quad (7)$$

The OLS estimates $\hat{\theta}_0$ and $\hat{\theta}_1$ are therefore defined as the parameter values that minimize the RSS:

$$(\hat{\theta}_0, \hat{\theta}_1) = \underset{\theta_0, \theta_1}{\operatorname{argmin}} \text{RSS}(\theta_0, \theta_1). \quad (8)$$

Once the model has been fitted, the RSS evaluated at the estimated parameters can equivalently be written in terms of the residuals:

$$\text{RSS}(\hat{\theta}_0, \hat{\theta}_1) = \sum_{i=1}^N e_i^2. \quad (9)$$

1.1.2 Parameter Estimation

To obtain the OLS estimates analytically, we minimize $\text{RSS}(\theta_0, \theta_1)$ with respect to both parameters. The critical points of the RSS satisfy

$$\nabla \text{RSS}(\theta_0, \theta_1) = \left(\frac{\partial \text{RSS}}{\partial \theta_0}, \frac{\partial \text{RSS}}{\partial \theta_1} \right) = (0, 0). \quad (10)$$

Estimation of the Intercept

Differentiating the RSS with respect to θ_0 gives

$$\begin{aligned} \frac{\partial \text{RSS}}{\partial \theta_0} &= \frac{\partial}{\partial \theta_0} \sum_{i=1}^N (y_i - \theta_0 - \theta_1 x_i)^2 \\ &= -2 \sum_{i=1}^N (y_i - \theta_0 - \theta_1 x_i). \end{aligned}$$

Setting this derivative equal to zero,

$$-2 \sum_{i=1}^N (y_i - \hat{\theta}_0 - \hat{\theta}_1 x_i) = 0.$$

Therefore,

$$\begin{aligned} \sum_{i=1}^N (y_i - \hat{\theta}_0 - \hat{\theta}_1 x_i) &= 0 \\ \sum_{i=1}^N y_i - \sum_{i=1}^N \hat{\theta}_0 - \hat{\theta}_1 \sum_{i=1}^N x_i &= 0 \\ \sum_{i=1}^N y_i - N\hat{\theta}_0 - \hat{\theta}_1 \sum_{i=1}^N x_i &= 0. \end{aligned}$$

Hence,

$$\begin{aligned} \sum_{i=1}^N y_i &= N\hat{\theta}_0 + \hat{\theta}_1 \sum_{i=1}^N x_i \\ N\hat{\theta}_0 &= \sum_{i=1}^N y_i - \hat{\theta}_1 \sum_{i=1}^N x_i \\ \hat{\theta}_0 &= \frac{1}{N} \sum_{i=1}^N y_i - \hat{\theta}_1 \frac{1}{N} \sum_{i=1}^N x_i. \end{aligned}$$

Using

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i, \quad \bar{y} = \frac{1}{N} \sum_{i=1}^N y_i,$$

we obtain

$$\hat{\theta}_0 = \bar{y} - \hat{\theta}_1 \bar{x}. \tag{11}$$

Thus, the fitted regression line passes through the point $(\bar{x}, \bar{y})$.

Estimation of the Slope

Differentiating the RSS with respect to θ_1 gives

$$\begin{aligned}\frac{\partial \text{RSS}}{\partial \theta_1} &= \frac{\partial}{\partial \theta_1} \sum_{i=1}^N (y_i - \theta_0 - \theta_1 x_i)^2 \\ &= -2 \sum_{i=1}^N (y_i - \theta_0 - \theta_1 x_i) x_i.\end{aligned}$$

Setting this derivative equal to zero,

$$-2 \sum_{i=1}^N (y_i - \hat{\theta}_0 - \hat{\theta}_1 x_i) x_i = 0.$$

Therefore,

$$\begin{aligned}\sum_{i=1}^N (y_i - \hat{\theta}_0 - \hat{\theta}_1 x_i) x_i &= 0 \\ \sum_{i=1}^N y_i x_i - \hat{\theta}_0 \sum_{i=1}^N x_i - \hat{\theta}_1 \sum_{i=1}^N x_i^2 &= 0.\end{aligned}$$

Hence,

$$\sum_{i=1}^N y_i x_i = \hat{\theta}_0 \sum_{i=1}^N x_i + \hat{\theta}_1 \sum_{i=1}^N x_i^2.$$

Substituting the expression for the intercept from Equation 11,

$$\begin{aligned}\sum_{i=1}^N y_i x_i &= (\bar{y} - \hat{\theta}_1 \bar{x}) \sum_{i=1}^N x_i + \hat{\theta}_1 \sum_{i=1}^N x_i^2 \\ &= \bar{y} \sum_{i=1}^N x_i - \hat{\theta}_1 \bar{x} \sum_{i=1}^N x_i + \hat{\theta}_1 \sum_{i=1}^N x_i^2.\end{aligned}$$

Since

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i,$$

we have

$$\bar{x} \sum_{i=1}^N x_i = \frac{\left(\sum_{i=1}^N x_i\right)^2}{N}.$$

Therefore,

$$\sum_{i=1}^N y_i x_i - \bar{y} \sum_{i=1}^N x_i = \hat{\theta}_1 \left[\sum_{i=1}^N x_i^2 - \frac{\left(\sum_{i=1}^N x_i\right)^2}{N} \right].$$

Equivalently, using

$$\bar{y} = \frac{1}{N} \sum_{i=1}^N y_i,$$

the left-hand side can be written as

$$\sum_{i=1}^N y_i x_i - \bar{y} \sum_{i=1}^N x_i = \sum_{i=1}^N y_i x_i - \frac{\sum_{i=1}^N y_i}{N} \sum_{i=1}^N x_i.$$

To express this quantity in centered form, consider

$$\begin{aligned} \sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y}) &= \sum_{i=1}^N (x_i y_i - x_i \bar{y} - \bar{x} y_i + \bar{x} \bar{y}) \\ &= \sum_{i=1}^N x_i y_i - \bar{y} \sum_{i=1}^N x_i - \bar{x} \sum_{i=1}^N y_i + N \bar{x} \bar{y}. \end{aligned}$$

Since

$$\sum_{i=1}^N x_i = N \bar{x}, \quad \sum_{i=1}^N y_i = N \bar{y},$$

it follows that

$$\begin{aligned}\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y}) &= \sum_{i=1}^N x_i y_i - N\bar{x}\bar{y} - N\bar{x}\bar{y} + N\bar{x}\bar{y} \\ &= \sum_{i=1}^N x_i y_i - N\bar{x}\bar{y}.\end{aligned}$$

Since

$$N\bar{x}\bar{y} = \bar{y} \sum_{i=1}^N x_i,$$

we obtain

$$\sum_{i=1}^N y_i x_i - \bar{y} \sum_{i=1}^N x_i = \sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y}).$$

Similarly, for the right-hand side,

$$\begin{aligned}\sum_{i=1}^N (x_i - \bar{x})^2 &= \sum_{i=1}^N (x_i^2 - 2x_i\bar{x} + \bar{x}^2) \\ &= \sum_{i=1}^N x_i^2 - 2\bar{x} \sum_{i=1}^N x_i + N\bar{x}^2 \\ &= \sum_{i=1}^N x_i^2 - 2N\bar{x}^2 + N\bar{x}^2 \\ &= \sum_{i=1}^N x_i^2 - N\bar{x}^2.\end{aligned}$$

Because

$$N\bar{x}^2 = \frac{\left(\sum_{i=1}^N x_i\right)^2}{N},$$

we have

$$\sum_{i=1}^N (x_i - \bar{x})^2 = \sum_{i=1}^N x_i^2 - \frac{\left(\sum_{i=1}^N x_i\right)^2}{N}.$$

The slope equation can therefore be written as

$$\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y}) = \hat{\theta}_1 \sum_{i=1}^N (x_i - \bar{x})^2.$$

Provided that the predictor is not constant, so that

$$\sum_{i=1}^N (x_i - \bar{x})^2 > 0,$$

the OLS estimate of the slope is

$$\hat{\theta}_1 = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})^2}. \quad (12)$$

If covariance and variance are defined using the same normalization, this can equivalently be expressed as

$$\hat{\theta}_1 = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}. \quad (13)$$

Together, Equation 11 and Equation 12 give the OLS parameter estimates for simple linear regression:

$$\begin{aligned} \hat{\theta}_0 &= \bar{y} - \hat{\theta}_1 \bar{x}, \\ \hat{\theta}_1 &= \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})^2}. \end{aligned} \quad (14)$$

1.1.3 Hessian and the Minimum of the RSS

The first-order conditions identify a critical point of the RSS. To determine the nature of this critical point, we examine the second derivatives of the objective function.

The second derivative with respect to θ_0 is

$$\begin{aligned}\frac{\partial^2 \text{RSS}}{\partial \theta_0^2} &= -2 \sum_{i=1}^N (-1) \\ &= 2N.\end{aligned}$$

The second derivative with respect to θ_1 is

$$\begin{aligned}\frac{\partial^2 \text{RSS}}{\partial \theta_1^2} &= -2 \sum_{i=1}^N (-x_i)x_i \\ &= 2 \sum_{i=1}^N x_i^2.\end{aligned}$$

The mixed second derivative is

$$\begin{aligned}\frac{\partial^2 \text{RSS}}{\partial \theta_0 \partial \theta_1} &= -2 \sum_{i=1}^N (-x_i) \\ &= 2 \sum_{i=1}^N x_i.\end{aligned}$$

Therefore, the Hessian matrix of the RSS is

$$H = \frac{\partial^2 \text{RSS}}{\partial \theta^2} = \begin{bmatrix} 2N & 2 \sum_{i=1}^N x_i \\ 2 \sum_{i=1}^N x_i & 2 \sum_{i=1}^N x_i^2 \end{bmatrix}. \quad (15)$$

The first leading principal minor is

$$2N > 0.$$

The determinant of the Hessian is

$$\begin{aligned}
\det(H) &= (2N) \left(2 \sum_{i=1}^N x_i^2 \right) - \left(2 \sum_{i=1}^N x_i \right)^2 \\
&= 4N \sum_{i=1}^N x_i^2 - 4 \left(\sum_{i=1}^N x_i \right)^2 \\
&= 4N \left[\sum_{i=1}^N x_i^2 - \frac{\left(\sum_{i=1}^N x_i \right)^2}{N} \right] \\
&= 4N \sum_{i=1}^N (x_i - \bar{x})^2.
\end{aligned} \tag{16}$$

If the predictor has non-zero variation,

$$\sum_{i=1}^N (x_i - \bar{x})^2 > 0,$$

then

$$\det(H) > 0.$$

Since the first leading principal minor is also positive, the Hessian is positive definite:

$$H \succ 0.$$

Consequently, $\text{RSS}(\theta_0, \theta_1)$ is a strictly convex function of the parameters. The critical point $(\hat{\theta}_0, \hat{\theta}_1)$ is therefore the unique global minimum of the RSS:

$$\text{RSS}(\hat{\theta}_0, \hat{\theta}_1) = \min_{\theta_0, \theta_1} \text{RSS}(\theta_0, \theta_1). \tag{17}$$

Thus, provided that the predictor has non-zero variance, the OLS estimates are uniquely determined and define the regression line that minimizes the sum of the squared residuals over the training data.

1.1.4 Python Implementation

The OLS estimates derived in Equation 14 can be implemented directly in Python. The following implementation computes these estimates from first principles, without relying on a regression library. A synthetic dataset is then generated from a known linear model so that the estimated regression line can be compared with the underlying data-generating process.

```
1 # Import libraries
2 import numpy as np
3 import matplotlib.pyplot as plt
4
5
6 def min_sqr(x, y):
7     """
8     Estimate the parameters of a simple linear regression model using
9     ordinary least squares.
10
11     Parameters
12     -----
13     x : array-like
14         Predictor values.
15     y : array-like
16         Observed response values.
17
18     Returns
19     -----
20     theta_0 : float
21         Estimated intercept of the regression line.
22     theta_1 : float
23         Estimated slope of the regression line.
24     """
25     x_bar = np.mean(x)
26     y_bar = np.mean(y)
27
28     theta_1 = (
29         np.sum((x - x_bar) * (y - y_bar))
30         / np.sum((x - x_bar) ** 2)
31     )
32     theta_0 = y_bar - theta_1 * x_bar
33
34     return theta_0, theta_1
```

```

35
36
37 # Random number generator
38 rng = np.random.default_rng(42)
39
40 # Number of observations
41 N = 1000
42
43 # True model parameters
44 theta_0_true = 1.0
45 theta_1_true = 2.0
46
47 # Generate predictor values
48 x = rng.normal(size=N)
49
50 # Generate random errors
51 epsilon = rng.normal(size=N)
52
53 # Generate responses from the linear model
54 y = theta_0_true + theta_1_true * x + epsilon
55
56 # Estimate regression coefficients
57 theta_0, theta_1 = min_sqr(x, y)
58
59 theta_0, theta_1

```

```
(np.float64(0.9183877310814965), np.float64(1.9919037171406144))
```

The estimated parameters should be close to the true values used to generate the data, with differences arising from the random error term and finite-sample variation. The estimated and true parameter values can be compared directly:

```

1 print(f"True intercept:   {theta_0_true:.4f}")
2 print(f"Estimated intercept: {theta_0:.4f}")
3 print(f"True slope:       {theta_1_true:.4f}")
4 print(f"Estimated slope:   {theta_1:.4f}")

```

```

True intercept:   1.0000
Estimated intercept: 0.9184
True slope:       2.0000

```

Estimated slope: 1.9919

The fitted regression function is

$$\hat{y}(x) = \hat{\theta}_0 + \hat{\theta}_1 x.$$

Using the estimated parameters, the fitted regression line can be evaluated over a sequence of predictor values and compared visually with the observed data.

```
1 # Predictor values used to draw the fitted regression line
2 x_seq = np.linspace(x.min(), x.max(), 200)
3
4 # Predicted response values
5 y_pred = theta_0 + theta_1 * x_seq
6
7 # Plot
8 plt.figure(figsize=(8, 6))
9
10 plt.scatter(x, y, alpha=0.5, label="Observed data")
11 plt.plot(x_seq, y_pred, label="OLS fitted line")
12
13 plt.title("Simple Linear Regression")
14 plt.xlabel("x")
15 plt.ylabel("y")
16
17 plt.grid(True)
18 plt.legend()
19
20 plt.show()
```

